Applied LLMs for Forecasting and Policy Research

Lecture 3: Downstream Regression

Stephen Hansen
University College London

BANK OF CANADA
BANQUE DU CANADA

Text-as-Data Setup in Economics

1. z: latent variable of economic interest, e.g. policy uncertainty
2. x: text data, e.g. newspapers
3. y: outcome data, e.g. aggregate output

Ideal approach is to model y as a function of z.

Typical approach is to (i) use x to create proxy measure z'; (ii) model y as a function of z'.

Applications to Forecasting

[Ellingsen et al., 2022]: topic time series from US newspapers has incremental forecasting value for US macro aggregates.

[Kalamara et al., 2022]: out-of-sample predictive value of newspaper content for UK macro aggregates.

[Shapiro et al., 2022]: compares alternative ways of building sentiment with validation data.

[Ashwin et al., 2024] and [Barbaglia et al., 2024]: nowcasting/forecasting eurozone GDP with sentiment.

Two Questions

1. How sensitive is downstream inference to choice of upstream model?
2. How is downstream inference affected by separation of (i) and (ii)? [Battaglia et al., 2025].

Choosing Among Algorithms

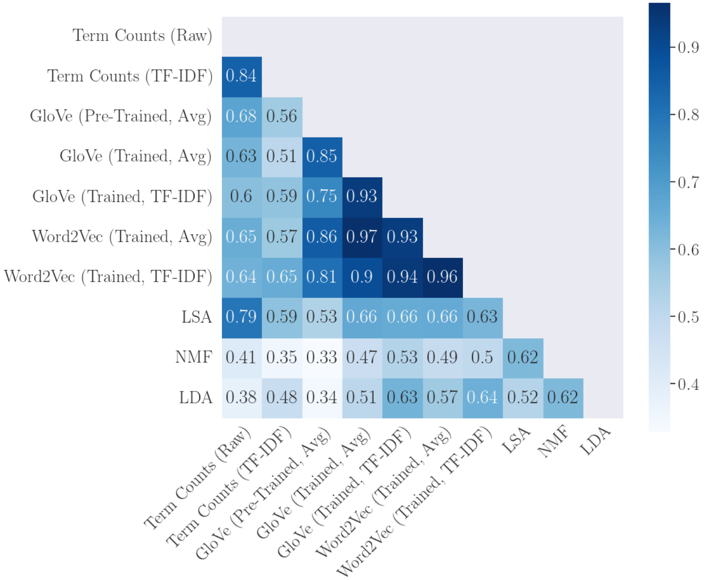
Multiple algorithms for document similarity: bag-of-words, word2vec, BERT, etc.

No clear metric of how to judge which is best and human labeling is hard.

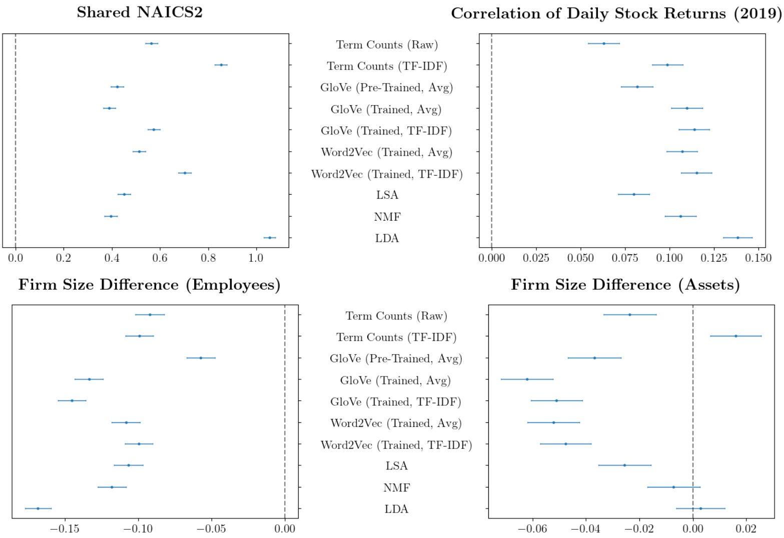
We compute document similarity in the context of 10-K risk factors using randomly sampled pairs from the universe of 2019 filing firms.

Keep data constant, and vary the algorithm used for similarity comparison.

Pearson Correlation Between Similarity Scores Across Pairs



Downstream Regression Results



Formal Setup from [Battaglia et al., 2025]

Want: perform inference on γ and/or α in the model

$$Y_i = \boldsymbol{\gamma}^T \boldsymbol{\theta}_i + \boldsymbol{\alpha}^T \mathbf{q}_i + \varepsilon_i, \qquad \mathbb{E}\left[\varepsilon_i \middle| \boldsymbol{\theta}_i, \mathbf{q}_i\right] = 0,$$

- $\boldsymbol{\theta}_i$ is a latent variable of economic interest
- $\mathbf{q}_i$ are observed numeric covariates
- Unstructured/high-dim dataset $\mathbf{x}_i$ available for estimating $\boldsymbol{\theta}_i$

Formal Setup from [Battaglia et al., 2025]

Want: perform inference on γ and/or $\boldsymbol{\alpha}$ in the model

$$Y_i = \boldsymbol{\gamma}^T \boldsymbol{\theta}_i + \boldsymbol{\alpha}^T \mathbf{q}_i + \varepsilon_i, \qquad \mathbb{E}\left[\varepsilon_i \middle| \boldsymbol{\theta}_i, \mathbf{q}_i\right] = 0,$$

- $\boldsymbol{\theta}_i$ is a latent variable of economic interest
- $\mathbf{q}_i$ are observed numeric covariates
- Unstructured/high-dim dataset $\mathbf{x}_i$ available for estimating $\boldsymbol{\theta}_i$

Two-Step Strategy:

1. Estimate $\hat{\boldsymbol{\theta}}_i$ of $\boldsymbol{\theta}_i$ obtained from unstructured data $\mathbf{x}_i$ and ML/AI model.
2. Regress Y_i on $\hat{\boldsymbol{\theta}}_i$ and $\mathbf{q}_i$. Perform inference treating $\hat{\boldsymbol{\theta}}_i$ as regular numeric data.

Valid Inference Without Validation Data

Recent work (mainly stats/poli sci) has pointed out potential for generated variables to cause problems

- General ML-generated variables: Fong and Tyler (2021), Allon et al. (2023), Angelopoulos et al. (2023a, 2023b), Zhang et al. (2023), Zrnic and Candès (2024), and Miao and Lu (2024)
- LLM-generated variables: Egami et al. (2023, 2024), Ludwig et al. (2025), Carlson and Dell (2025).

Valid Inference Without Validation Data

Recent work (mainly stats/poli sci) has pointed out potential for generated variables to cause problems

- General ML-generated variables: Fong and Tyler (2021), Allon et al. (2023), Angelopoulos et al. (2023a, 2023b), Zhang et al. (2023), Zrnic and Candès (2024), and Miao and Lu (2024)
- LLM-generated variables: Egami et al. (2023, 2024), Ludwig et al. (2025), Carlson and Dell (2025).

Papers propose bias corrections assuming a validation subsample in which $(Y_i, \boldsymbol{\theta}_i, \hat{\boldsymbol{\theta}}_i)$ are observed

- Inference valid when size of labeled/unlabeled data approximately equal.
- Typically requires human labelers to construct validation set

  But use of ML/AI typically motivated by large cost of labeling.
- In economic data, unclear a true label is observed (sentiment, uncertainty).

Valid Inference Without Validation Data

Recent work (mainly stats/poli sci) has pointed out potential for generated variables to cause problems

- General ML-generated variables: Fong and Tyler (2021), Allon et al. (2023), Angelopoulos et al. (2023a, 2023b), Zhang et al. (2023), Zrnic and Candès (2024), and Miao and Lu (2024)
- LLM-generated variables: Egami et al. (2023, 2024), Ludwig et al. (2025), Carlson and Dell (2025).

Papers propose bias corrections assuming a validation subsample in which $(Y_i, \boldsymbol{\theta}_i, \hat{\boldsymbol{\theta}}_i)$ are observed

- Inference valid when size of labeled/unlabeled data approximately equal.
- Typically requires human labelers to construct validation set

  But use of ML/AI typically motivated by large cost of labeling.
- In economic data, unclear a true label is observed (sentiment, uncertainty).

Need for valid inference without validation data.

This Paper

1. Asymptotic framework where measurement error $\downarrow$ as sample size $\uparrow$.

   Two-step CIs have right width but wrong centering (bias) which depends on relative importance of

   (a) measurement error in $\hat{\theta}_i$
   (b) sampling error in downstream model

   Valid two-step inference requires (a) $\ll$ (b)

   This is not the case in most leading applications

2. Two solutions: bias correction + one-step strategy
   NB: Measurement error in AI/ML-generated variables is non-classical.

3. Shows empirical relevance in several empirical applications

Example 1: AI/ML-Generated Labels

- Leading use case: missing θ_i is binary (e.g., race indicator): Goldsmith-Pinkham and Shue (2023), Adams-Prassl et. al. (2023), Argyle et al. (2025), and Wu and Yang (2024)
- Generate estimate $\hat{\theta}_i$ of θ_i using unstructured data $\mathbf{x}_i$ (e.g., voter registration data)
- Regress Y_i on $\hat{\theta}_i$ and controls $\mathbf{q}_i$

Example 1: AI/ML-Generated Labels

- Leading use case: missing θ_i is binary (e.g., race indicator): Goldsmith-Pinkham and Shue (2023), Adams-Prassl et. al. (2023), Argyle et al. (2025), and Wu and Yang (2024)
- Generate estimate $\hat{\theta}_i$ of θ_i using unstructured data $\mathbf{x}_i$ (e.g., voter registration data)
- Regress Y_i on $\hat{\theta}_i$ and controls $\mathbf{q}_i$
- Measurement error due to misclassification error:

$$\Pr(\theta_i = 1 | \mathbf{x}_i, \mathbf{q}_i) \neq \Pr(\hat{\theta}_i = 1 | \mathbf{x}_i, \mathbf{q}_i)$$

Example 2: Indices

- Several influential works generate indices by classifying documents + aggregating: Baker Bloom Davis (2016), Caldara and Iacoviello (2022), Gorodnichenko Pham Talavera (2023).
- Each month observe C_i documents (e.g., set of newspapers)
- Of these, X_i are classified as pertaining to concept (e.g., policy uncertainty)
- Latent true uncertainty $\theta_i \in [0, 1]$
- Naive estimator: $\hat{\theta}_i = X_i / C_i$ (cf. BBD's EPU measure)

Example 2: Indices

- Several influential works generate indices by classifying documents + aggregating: Baker Bloom Davis (2016), Caldara and Iacoviello (2022), Gorodnichenko Pham Talavera (2023).
- Each month observe C_i documents (e.g., set of newspapers)
- Of these, X_i are classified as pertaining to concept (e.g., policy uncertainty)
- Latent true uncertainty $\theta_i \in [0, 1]$
- Naive estimator: $\hat{\theta}_i = X_i / C_i$ (cf. BBD's EPU measure)
- Natural model is $X_i | C_i, \theta_i \sim \text{Binomial}(C_i, \theta_i \beta_1 + (1 - \theta_i)\beta_0)$ where β_x is the probability that a document with true label x is classified a one.
- Measurement error in $\hat{\theta}_i$ arises from misclassification error ($\boldsymbol{\beta}$) and sampling error (C_i).

Example 2: Indices — Simulation Calibrated to Gorodnichenko et al. (2023)

	Bias			RMdSE			Coverage		
Configuration	1	2	3	1	2	3	1	2	3
				$n = 200$					
2-Step	-0.433	-0.218	-0.037	0.048	0.025	0.018	0.378	0.824	0.931
Joint	-0.003	0.007	0.004	0.024	0.020	0.018	0.945	0.948	0.938
				$n = 800$					
2-Step	-0.215	-0.041	0.084	0.024	0.010	0.012	0.507	0.942	0.894
Joint	0.004	-0.006	-0.006	0.011	0.010	0.010	0.956	0.950	0.950
				$n = 3200$					
2-Step	-0.042	0.085	0.158	0.006	0.009	0.017	0.887	0.739	0.353
Joint	-0.005	-0.002	-0.003	0.005	0.005	0.005	0.942	0.941	0.943

Asymptotics: General Case

- Consider a sequence of DGPs for $(Y_i, \boldsymbol{\theta}_i, \hat{\boldsymbol{\theta}}_i, \mathbf{q}_i, \mathbf{x}_i)_{i=1}^n$ indexed by sample size n, in which
$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \hat{\boldsymbol{\theta}}_i (\hat{\boldsymbol{\theta}}_i - \boldsymbol{\theta}_i)^T \rightarrow_p \kappa \, \boldsymbol{\Omega},$$
(expressions are DGP-specific)
- Scalar $\kappa \geq 0$ measures the importance of measurement error relative to sampling error
- Positive κ allows both sampling error and measurement error to play a role

Asymptotics: General Case

- Consider a sequence of DGPs for $(Y_i, \boldsymbol{\theta}_i, \hat{\boldsymbol{\theta}}_i, \mathbf{q}_i, \mathbf{x}_i)_{i=1}^n$ indexed by sample size n, in which

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \hat{\boldsymbol{\theta}}_i (\hat{\boldsymbol{\theta}}_i - \boldsymbol{\theta}_i)^T \to_p \kappa \, \boldsymbol{\Omega},$$

  (expressions are DGP-specific)
- Scalar $\kappa \geq 0$ measures the importance of measurement error relative to sampling error
- Positive κ allows both sampling error and measurement error to play a role
- Example 1 expression is

$$\sqrt{n} \times \underbrace{\mathbb{E}\left[\hat{\theta}_i (1 - \theta_i)\right]}_{\text{false-positive rate}} \to \kappa, \qquad \boldsymbol{\Omega} = 1$$

- Reflects prevailing trend: increasingly large data sets + increasingly accurate algorithms

Theorem on Two-Step Inference

Theorem: Two-Step Inference is Invalid Unless $\kappa = 0$

1. OLS estimator $\hat{\boldsymbol{\psi}} = (\hat{\boldsymbol{\gamma}}, \hat{\boldsymbol{\alpha}})$ of $\boldsymbol{\psi} = (\boldsymbol{\gamma}, \boldsymbol{\alpha})$ from regressing Y_i on $\hat{\boldsymbol{\xi}}_i = (\hat{\boldsymbol{\theta}}_i, \mathbf{q}_i)$ has asy dist

$$\sqrt{n}\left(\hat{\boldsymbol{\psi}} - \boldsymbol{\psi}\right) \to_d N\left(-\kappa \mathbb{E}[\boldsymbol{\xi}_i \boldsymbol{\xi}_i^T]^{-1} \begin{pmatrix} \boldsymbol{\Omega} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \boldsymbol{\psi}\,,\ \underbrace{\mathbb{E}[\boldsymbol{\xi}_i \boldsymbol{\xi}_i^T]^{-1} \mathbb{E}[\varepsilon_i^2 \boldsymbol{\xi}_i \boldsymbol{\xi}_i^T] \mathbb{E}[\boldsymbol{\xi}_i \boldsymbol{\xi}_i^T]^{-1}}_{=:\mathbf{V}}\right)$$

where $\boldsymbol{\xi}_i = (\boldsymbol{\theta}_i, \mathbf{q}_i)$ are the "true" covariates

Theorem on Two-Step Inference

Theorem: Two-Step Inference is Invalid Unless $\kappa = 0$

1. OLS estimator $\hat{\boldsymbol{\psi}} = (\hat{\boldsymbol{\gamma}}, \hat{\boldsymbol{\alpha}})$ of $\boldsymbol{\psi} = (\boldsymbol{\gamma}, \boldsymbol{\alpha})$ from regressing Y_i on $\hat{\boldsymbol{\xi}}_i = (\hat{\boldsymbol{\theta}}_i, \mathbf{q}_i)$ has asy dist

$$\sqrt{n}\left(\hat{\boldsymbol{\psi}} - \boldsymbol{\psi}\right) \to_d N\left(-\kappa \mathbb{E}[\boldsymbol{\xi}_i \boldsymbol{\xi}_i^T]^{-1} \begin{pmatrix} \boldsymbol{\Omega} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \boldsymbol{\psi}\,,\ \underbrace{\mathbb{E}[\boldsymbol{\xi}_i \boldsymbol{\xi}_i^T]^{-1} \mathbb{E}[\varepsilon_i^2 \boldsymbol{\xi}_i \boldsymbol{\xi}_i^T] \mathbb{E}[\boldsymbol{\xi}_i \boldsymbol{\xi}_i^T]^{-1}}_{=:\mathbf{V}}\right)$$

   where $\boldsymbol{\xi}_i = (\boldsymbol{\theta}_i, \mathbf{q}_i)$ are the "true" covariates

2. Eicker–Huber–White standard errors are consistent for all $\kappa \geq 0$:

$$\hat{\mathbf{V}} := \left(\frac{1}{n}\sum_{i=1}^{n} \hat{\boldsymbol{\xi}}_i \hat{\boldsymbol{\xi}}_i^T\right)^{-1} \left(\frac{1}{n}\sum_{i=1}^{n} \hat{\varepsilon}_i^2 \hat{\boldsymbol{\xi}}_i \hat{\boldsymbol{\xi}}_i^T\right) \left(\frac{1}{n}\sum_{i=1}^{n} \hat{\boldsymbol{\xi}}_i \hat{\boldsymbol{\xi}}_i^T\right)^{-1} \to_p \mathbf{V}$$

How to Do Valid Inference

1. **Explicit Bias Correction**: use analytical expressions in Theorem to adjust two-step estimates/CIs

   Advantage: Simple and scalable

   Disadvantage: Not feasible in complex models; poor approximation with large κ

2. **Joint Estimation**: MLE using joint likelihood for upstream IR model + regression model

   Advantage: General purpose and flexible

   Disadvantage: More computationally demanding

References I

Ashwin, J., Kalamara, E., and Saiz, L. (2024).
Nowcasting Euro area GDP with news sentiment: A tale of two crises.
Journal of Applied Econometrics, 39(5):887–905.

Barbaglia, L., Consoli, S., and Manzan, S. (2024).
Forecasting GDP in Europe with textual data.
Journal of Applied Econometrics, 39(2):338–355.

Battaglia, L., Christensen, T., Hansen, S., and Sacher, S. (2025).
Inference for Regression with Variables Generated by AI or Machine Learning.

Ellingsen, J., Larsen, V. H., and Thorsrud, L. A. (2022).
News media versus FRED-MD for macroeconomic forecasting.
Journal of Applied Econometrics, 37(1):63–81.

Kalamara, E., Turrell, A., Redl, C., Kapetanios, G., and Kapadia, S. (2022).
Making text count: Economic forecasting using newspaper text.
Journal of Applied Econometrics, 37(5):896–919.

References II

Shapiro, A. H., Sudhof, M., and Wilson, D. J. (2022).
Measuring news sentiment.
Journal of Econometrics, 228(2):221–243.